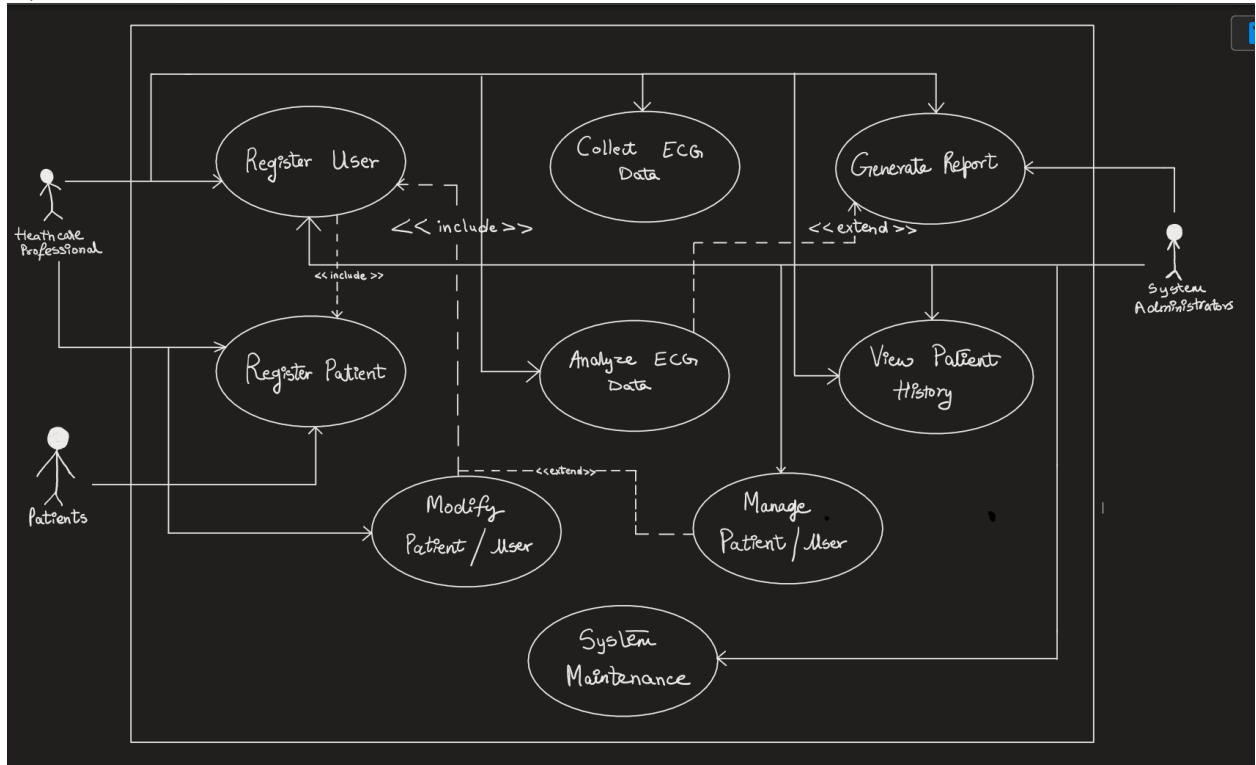


CE202 SOFTWARE ENGINEERING ASSIGNMENT 1

CARDIAC DIAGNOSIS SYSTEMS

1. A)



B) Register User (Healthcare Professionals and System Admins)

I) Actors – Healthcare Professionals, System Administration

II) Preconditions – the user should have necessary information, and the user should be a Healthcare Professional or a System Administration; also, not present in the system

III) Postconditions – logged in the system with account made successfully

IV) Purpose – to allow new users to register to access system functionalities

V) Description – the use case allows a user to set up an account by entering their information (name, email, password, and position), verifying the provided information and establishes a new user profile with the designated role

VI) Alternative Courses – if the email address is linked to an existing user, the system will show an error message and ask the user to provide a different email; if the password fails to satisfy the system's security standards (such as minimum length and character variety), the system will ask the user to modify the password

VII) Errors – duplicate email id, invalid format of mail id, password and contact information

Register Patient

- I) Actors – Healthcare Professionals
- II) Preconditions – logged into the system so the user has necessary permissions, allowing to store patient information
- III) Postconditions – new patient account is created associating it with a healthcare provider, with having details and entries made about the patient in the database
- IV) Purpose – creating, storing, management, ECG data collection and diagnosis
- V) Description – entering patients' personal information, verifying them in system and creating a new patient profile corresponding with a healthcare provider
- VI) Alternative Courses – if the entry is incomplete or invalid information, the system will prompt to correct the details before going ahead
- VII) Errors – duplicate patient entry, invalid or missing patient personal information

Collect ECG Data

- I) Actors – Healthcare Professionals
- II) Preconditions – patient profile should exist and the healthcare provider should have an access to the ECG monitor device
- III) Postconditions – data should be successfully recorded, stored, and correspond with the selected patient profile
- IV) Purpose – to collect, analyze ECG data for diagnosis
- V) Description – ECG monitor device connected to the patient to record and transmit the data into a digital format for future analysis and reporting
- VI) Alternative Courses – device fails to collect data which notifies the healthcare provider and prompts for a retry
- VII) Errors – device malfunction, data transmission failure

Analyze ECG Data

- I) Actors – Healthcare Professionals
- II) Preconditions – the healthcare provider has access to the ECG data records which should be stored in the system for analysis
- III) Postconditions – the data is processed and analyzed by the system, and the results are returned to the healthcare personnel, the system would have identified any abnormalities related to the ECG data
- IV) Purpose – process, analyze, detect abnormalities and aid in diagnosis
- V) Description – the analysis engine analyzes the selected ECG data from the patients profile for any anomalies; the results is then returned to the healthcare provider
- VI) Alternative Courses – insufficient data for analysis which notified to the healthcare provider prompting to re-collect the ECG data and if no abnormalities are detected, the system will provide the report without any issues
- VII) Errors – corrupt data and system failure in processing the ECG data

Generate Report

- I) Actors – Healthcare Professionals, System Administrations
- II) Preconditions – the ECG data should be analyzed completely

- III) Postconditions – the report is generated and saved in the patients profile
- IV) Purpose – to create detailed report based on the analysis
- V) Description – to generate a report containing ECG analysis results with basic patients details and with any findings that can be used for further analysis
- VI) Alternative Courses – report generated gets corrupted and fails which logs the system for troubleshooting
- VII) Errors – the incomplete data causes failure in report generation

View Patient History

- I) Actors – Healthcare Professional
- II) Preconditions – the patients records must exist in the system
- III) Postconditions – history will be displayed on the interface
- IV) Purpose – to review patients' past data for further diagnosis and reference
- V) Description – with the support of the chronological history of the patient, the ECG data can be further analyzed
- VI) Alternative Courses – no records found
- VII) Errors – data retrieval failure

C) I) Healthcare Professional – Entity

Role – manages records of patients, collects and analyzes ECG data, and request to generate reports

II) Patient – Entity

Role – represents information of patient with personal details and medical history

III) System Administrator – Entity / Control

Role – manages user accounts, permissions, and system maintenance tasks

IV) ECG Data – Entity

Role – ECG data is held and correlated to the patient records

V) Report – Entity

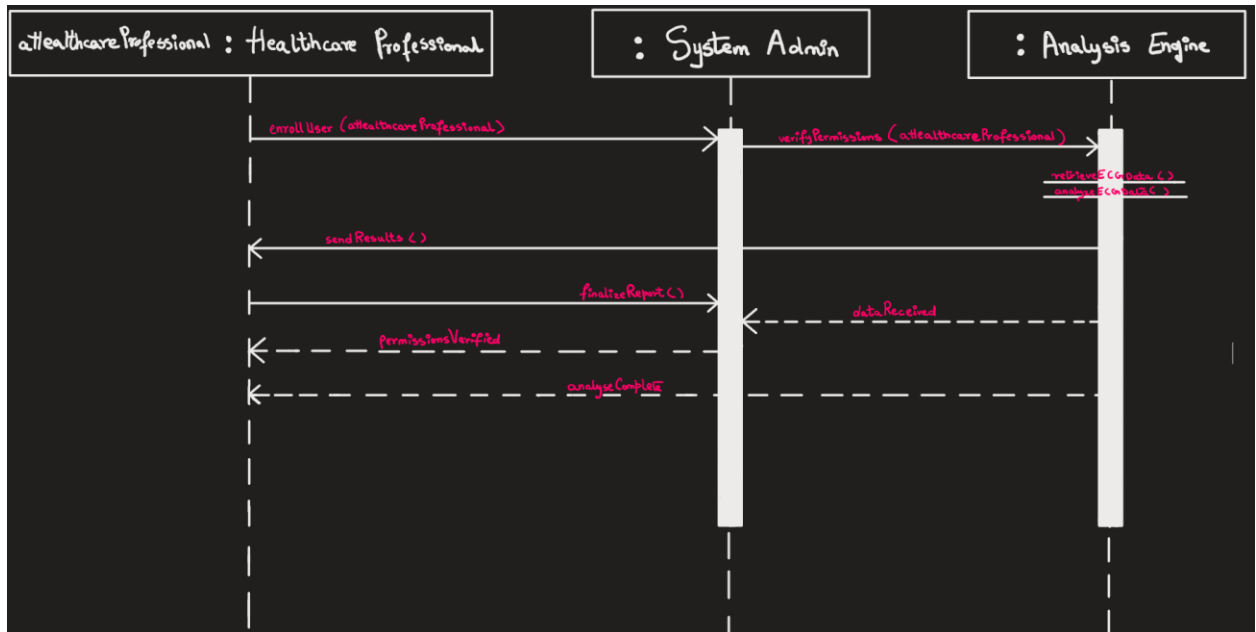
Role – contains analysis findings and diagnostic information for patient history

VI) User Registration – Boundary

Role – managing user registration process

VII) Patient Registration – Boundary

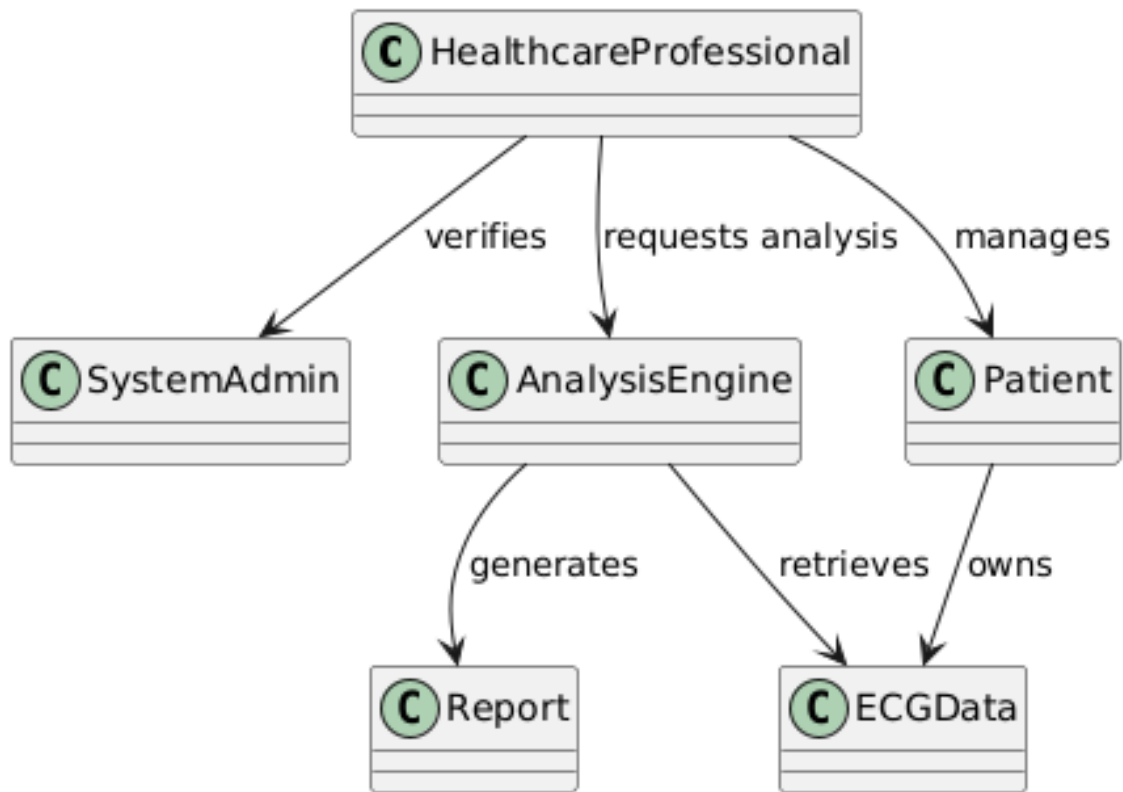
Role – patient registration managed and linking to the patient profile



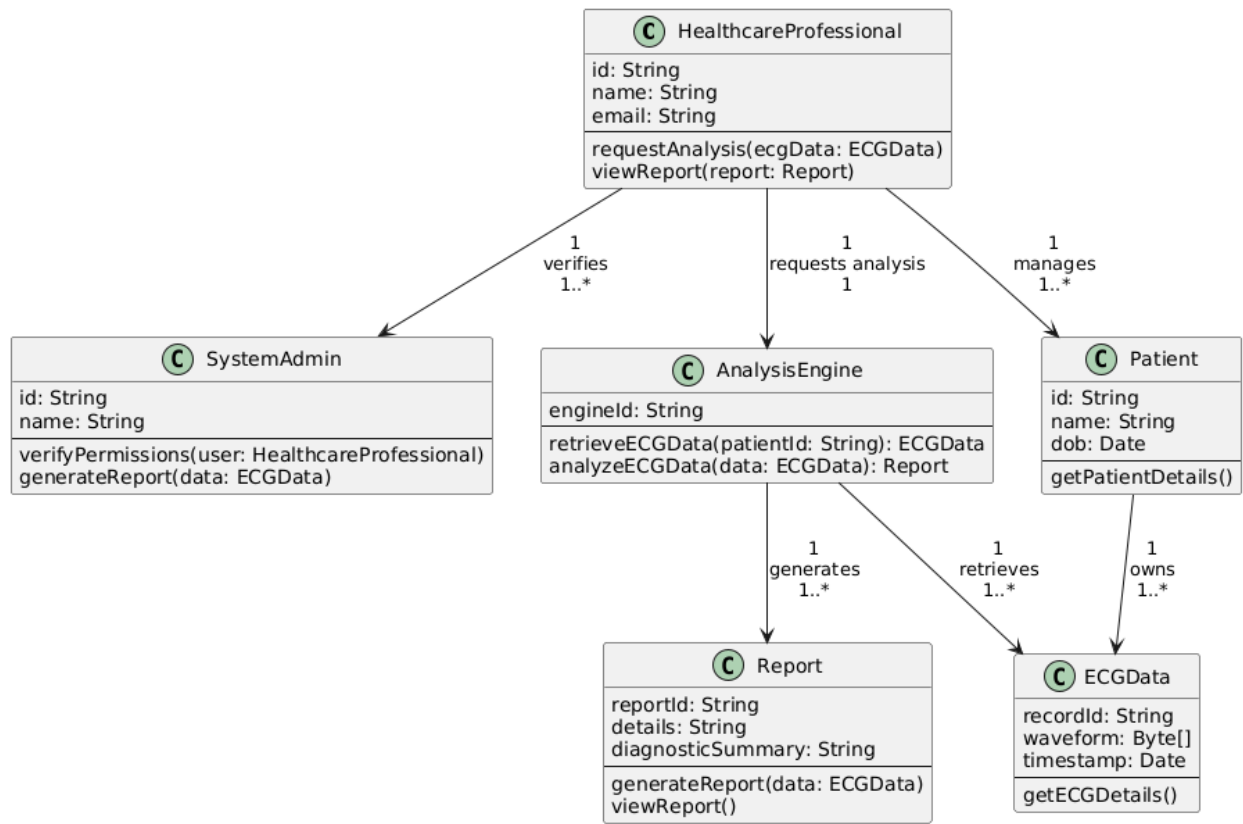
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Class	Responsibilities	Collaborations
Analysis Engine	To retrieve and analyze the ECG data, to detect abnormalities and to send results	Healthcare Professional and System Admin
Healthcare Professional	Initial analysis, get results and request the generation of specific reports	System Admin and Analysis Engine
System Admin	Verifying permissions, managing reports and handling security of the system	Healthcare Professional and Analysis Engine

4. A)



B)



C) The sequence and class diagrams exhibit a high degree of consistency with one another, as well as with the requirements analysis, effectively encapsulating the "Analyze ECG Data" use case. The sequence diagram delineates the flow of messages and interactions among the primary participants: HealthcareProfessional, SystemAdmin, and AnalysisEngine. It aligns closely with the requirements by incorporating essential steps such as permission verification, ECG data retrieval, data analysis, and report generation. These interactions are reflected in the class diagram, which delineates the fundamental entities, their attributes, and their interrelations.

The class diagram articulates the principal entities within the system, including HealthcareProfessional, SystemAdmin, AnalysisEngine, ECGData, Report, and Patient. It illustrates the associations among these classes (for example, HealthcareProfessional oversees Patient and solicits analysis from AnalysisEngine), which correspond to the interactions depicted in the sequence diagram. Moreover, operations within the class diagram, such as requestAnalysis() in HealthcareProfessional and analyzeECGData() in AnalysisEngine, directly align with the messages represented in the sequence diagram, thereby ensuring coherence between the two.

In compliance with the specified requirements, several additional attributes and operations were incorporated to enhance the system's robustness. For instance, unique identifiers (id) were introduced in classes such as HealthcareProfessional, SystemAdmin, and Patient to facilitate accurate tracking of entities. Attributes like recordId and timestamp in ECGData

were added to manage metadata associated with ECG records, while `reportId` and `diagnosticSummary` were included in `Report` to effectively document the results of analyses.

New operations, including `retrieveECGData()` and `analyzeECGData()` in `AnalysisEngine`, as well as `generateReport()` in `Report`, were implemented to mirror the functionalities of a real-world system. These operations are essential for tasks such as retrieving ECG data specific to patients, processing that data, and generating diagnostic reports. The integration of these features ensures that the system is comprehensive and functional.